

DIFFERENTIAL EQUATIONS

Lecture 1 — Introduction to Differential Equations

Branches of Calculus

Calculus is divided into two main branches, each dealing with a fundamentally different operation on functions.

Differential Calculus	Integral Calculus
Differential: an independent (infinitesimally small) change in a variable. Derivative: the rate of change of one variable with respect to another — i.e. a dependent change. The process of finding derivatives is called differentiation.	- Find the area under a curve. - Find the original function from its derivative (anti-differentiation). - Deals with accumulation of quantities.

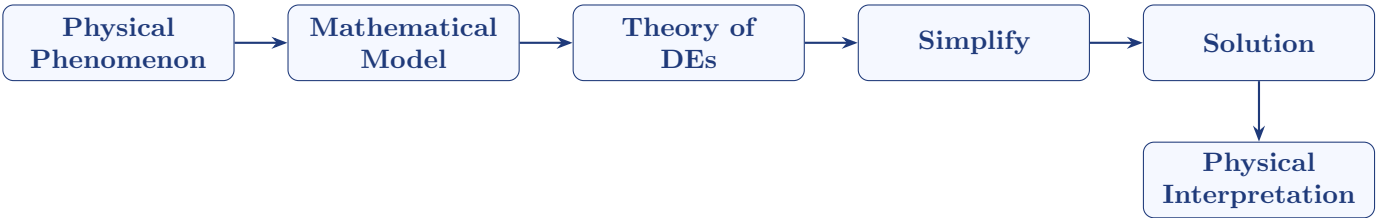
Key distinction: A *differential* refers to a tiny independent change in a variable (like dx). A *derivative* dy/dx is the ratio of the dependent change dy to the independent change dx — it measures how fast y responds to a change in x .

From Physical Phenomena to Differential Equations

The Modelling Pipeline

Most real-world phenomena — especially those involving change over time or space — cannot be described by simple algebraic equations. Instead, they are described by equations involving **rates of change**, i.e. derivatives. This is the central motivation for studying differential equations.

The journey from a physical observation to a usable solution follows a structured pipeline:



Definition: Differential Equation (Informal)

A **differential equation (DE)** is a mathematical equation that relates a function to its derivatives. It arises naturally whenever we model a quantity that changes continuously — such as velocity, temperature, population, or electric current.

The Two Key Steps

When faced with a physical problem that leads to a DE, always remember the two core steps:

Step	Name	What it means
1	IDENTIFY	Recognise what type of DE you are dealing with (order, degree, ODE vs PDE, linear vs non-linear)
2	SIMPLIFY	Reduce the DE to a standard form and apply the appropriate solution method

DEs provide **short-term local information** about a system — they describe how the system is *changing* at any given instant, not necessarily where it will be in the long run. The solution of a DE then gives the global behaviour.

Physical Examples that Lead to DEs

The following are real phenomena that are modelled by DEs. In each case, the physical law translates directly into a relationship involving a derivative.

Phenomenon	Physical Law	Resulting DE
Newton's Law of Cooling	Rate of heat loss $\propto$ temperature difference	$\frac{dT}{dt} = -k(T - T_{\text{env}})$
Radioactive Decay	Rate of decay $\propto$ amount present	$\frac{dN}{dt} = -\lambda N$
Simple Harmonic Motion	Restoring force $\propto$ displacement	$\frac{d^2x}{dt^2} + \omega^2 x = 0$
Population Growth	Growth rate $\propto$ population size	$\frac{dP}{dt} = kP$
Electric Circuits (RL)	Kirchhoff's Voltage Law	$L \frac{dI}{dt} + RI = V(t)$

ODE vs PDE

Ordinary Differential Equations

Definition: Ordinary Differential Equation (ODE)

A DE that involves **ordinary derivatives** of the dependent variable with respect to **exactly one** independent variable. It models **one-dimensional** phenomena.

Example: $\frac{dy}{dx} + y \cos x = \sin x$ (y depends only on x)

Partial Differential Equations

Definition: Partial Differential Equation (PDE)

A DE that involves **partial derivatives** of the dependent variable with respect to **more than one** independent variable. It models **multi-dimensional** phenomena.

Example: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ (u depends on x, y, z)

Comparison Table

ODE	PDE
Contains ordinary derivatives $\frac{d}{dx}$	Contains partial derivatives $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \dots$
One independent variable	More than one independent variable
Models 1-D phenomena (e.g. motion along a line, decay over time)	Models n-D phenomena (e.g. heat in a plate, waves in space)
Examples: $y' + y = 0$, $y'' + \omega^2 y = 0$	Examples: $u_{xx} + u_{yy} = 0$ (Laplace), $u_t = k u_{xx}$ (Heat)

In this course we focus primarily on **ODEs**. PDEs appear briefly as examples to illustrate the definition and the distinction, but their solution methods are covered in a separate course.

Summary

Concept	Key Point
Differential Equation	Relates a function to its derivatives; arises from modelling change
Modelling Pipeline	Physical Phenomenon $\rightarrow$ Model $\rightarrow$ DE $\rightarrow$ Simplify $\rightarrow$ Solution $\rightarrow$ Interpretation
Two Core Steps	IDENTIFY the type, then SIMPLIFY to a standard form
ODE	One independent variable; ordinary derivatives; 1-D phenomena
PDE	More than one independent variable; partial derivatives; n -D phenomena